

Degree-one optimal polynomial approximants can have zeros arbitrarily close to the origin

Candidate full solution to the zero-depth question in arXiv:2405.16943 and 2508.15938

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Status: candidate full solution, likely valid. For each $s \in \{1, 2, \infty\}$ and each $\varepsilon > 0$, there is a polynomial f , nonvanishing on the closed bidisk, whose degree-one optimal polynomial approximant p to $1/f$ has

$$\inf\{\|z\|_s : p(z) = 0\} < \varepsilon.$$

Thus the sharp universal zero-exclusion radius is zero for all three norms. The construction concentrates $|f|^2$ near two almost co-rotating points of the distinguished boundary; the OPA normal equations then converge to a two-point interpolation system with coefficients of order m .

1 The question and the later numerical benchmark

Bénéteau, Khavinson, and Seco ask how far inside the bidisk zeros of an optimal polynomial approximant (OPA) can get, explicitly distinguishing the ℓ_1 , ℓ_2 , and ℓ_∞ distances in $\mathbb{C}^2$ [1, pp. 13–14]. Their complete statement is reproduced on the next two pages.

j	Numer. q_j	Denom. q_j	$q_j \approx$	$H_j \in$	$H_j/(j+1) \in$
0	.	.	.	[0.500, 0.500]	[0.500, 0.500]
1	25	8	3.125	[1.562, 1.563]	[0.781, 0.782]
2	245	144	1.701	[2.656, 2.661]	[0.885, 0.887]
3	21	16	1.312	[3.484, 3.494]	[0.871, 0.874]
4	363	320	1.134	[3.950, 3.966]	[0.790, 0.794]
5	1859	1800	1.032	[4.076, 4.097]	[0.679, 0.683]
6	325	336	0.967	[3.941, 3.966]	[0.563, 0.567]
7	1445	1568	0.921	[3.629, 3.657]	[0.453, 0.458]
8	6137	6912	0.887	[3.218, 3.248]	[0.357, 0.361]
9	931	1080	0.862	[2.773, 2.804]	[0.277, 0.281]
10	3703	4400	0.841	[2.332, 2.361]	[0.212, 0.215]
11	14375	17424	0.825	[1.923, 1.951]	[0.160, 0.163]
12	675	832	0.811	[1.559, 1.585]	[0.119, 0.122]
13	7569	9464	0.799	[1.245, 1.268]	[0.088, 0.091]
14	27869	35280	0.789	[0.982, 1.002]	[0.065, 0.067]
15	3751	4800	0.781	[0.766, 0.784]	[0.047, 0.049]
16	13475	17408	0.774	[0.592, 0.608]	[0.034, 0.036]
17	47915	62424	0.767	[0.454, 0.467]	[0.025, 0.026]
18	6253	8208	0.761	[0.345, 0.356]	[0.018, 0.019]
19	21853	28880	0.756	[0.260, 0.270]	[0.013, 0.014]
20	75809	100800	0.752	[0.195, 0.204]	[0.009, 0.010]
21	3225	4312	0.747	[0.145, 0.153]	[0.006, 0.007]
22	33135	44528	0.744	[0.107, 0.114]	[0.004, 0.005]
23	112847	152352	0.740	[0.079, 0.085]	[0.003, 0.004]
24	14161	19200	0.737	[0.058, 0.063]	[0.002, 0.003]
25	47753	65000	0.734	[0.042, 0.047]	[0.001, 0.002]

5. FURTHER WORK

Here we want to emphasize that our refutation of the Weak Shanks Conjecture only opens the door to a very rich collection of important problems waiting to be investigated. For instance, even though we have proved that the zero sets of the OPAs for our function f do get inside the bidisk, we don't know how close to the origin the complex line of zeros for p_1 gets nor whether the OPAs of higher degree for that function have zeros in the bidisk. More generally, we don't know *how far* inside the bidisk zeros of any OPA can get. It would be interesting to study that extremal problem, as we did for one variable in [7], as we vary through all orderings of polynomials and/or through all functions in $H^2(\mathbb{D}^2)$. However, we cannot stress enough the great difference between the one and two variable problems there is as a result of the lack of a Fundamental Theorem of Algebra: the fact that zeros are not isolated means that the problem does not reduce to studying any sort of low dimensional polynomials. Moreover, one needs to choose what

“how far” means in this context: is this the point on the zero set of minimal ℓ_1 , ℓ_2 , or ℓ_∞ norm in $\mathbb{C}^2$?

On the other hand, the embedding of functions depending only on $z_1 + z_2$ proved useful in our study, partly because there is a natural version of the shift that creates a set of mutually orthogonal classes of monomials. Perhaps other closed subspaces can take the zeros further inside the disk, but it is not evident to us which spaces one can use.

One could also consider finding counterexamples with other orderings of monomials. If we choose an ordering in which 1, z_1 and z_2 are not the first 3 elements of the basis, or an ordering which does yield a symmetric OPA, we may get a different OPA, even for our function f . Is there any particular choice of ordering of the monomials for which the sequence of OPAs must be zero-free? It seems unlikely but far from our current reach.

What happens in higher dimensional polydisks? We do know that our result implies a similar failure of OPAs to be zero-free in higher dimensional polydisk Hardy spaces, from embedding remarks in [6] but do the answers to any of the other questions we pose vary in higher dimensions?

Perhaps most interesting, the phenomenon observed here (OPAs having zeros inside the domain for zero-free functions) can also be found in subspaces of $H^2(\mathbb{D}^2)$ called Dirichlet-type spaces $D_\alpha(\mathbb{D}^2)$. These spaces are defined in a similar fashion to $H^2(\mathbb{D}^2)$ but the norm of $z_1^j z_2^k$ is given by $(j+1)^{\alpha/2} (k+1)^{\alpha/2}$. For a fixed f , the inner products $\langle z_1^j z_2^k f, z_1^l z_2^n f \rangle_\alpha$ all vary continuously in terms of the parameter α . Hence for $\alpha > 0$ small enough, OPAs for f in $D_\alpha(\mathbb{D}^2)$ will have zeros inside $\mathbb{D}^2$. How far into the positive α values can we go? Is there a finite supremum for the values of α for which this phenomenon occurs?

What happens in the Hardy space over the unit ball of $\mathbb{C}^n$? Is the extremal for the symmetric functions space we obtained actually zero-free?

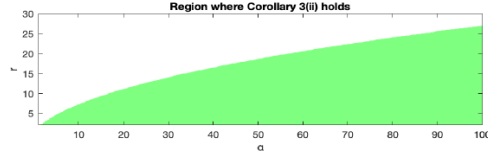
Finally, in many Dirichlet-type spaces in $\mathbb{C}$, the zeros of OPAs eventually escape every compact subset of the unit disk, due to the connection between OPAs and reproducing kernels in weighted spaces (see [7]). Does this behavior prevail in the bidisk? Along the same lines, does Jentzsch’s phenomenon related to accumulation of zeros of OPAs proven in [7] extend to higher dimensions? We hope to return to some of these questions in the future.

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Geronimo and Woerdeman later optimize numerically within the two-parameter family

$$f(z_1, z_2) = \left(1 - \frac{z_1 + z_2}{r}\right)^{-\alpha}$$

and report the value 0.975766335 [2, p. 8]. Their complete source page is reproduced below. The theorem here shows that this benchmark is a feature of that family, not a positive universal barrier.



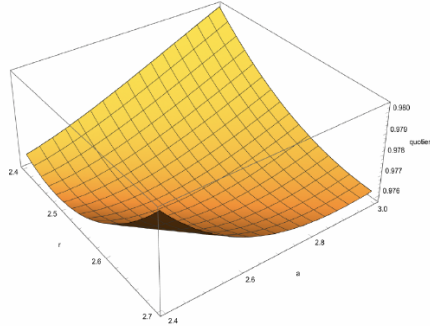
In the last section of [3] the question was asked how close to 0 the roots of a degree one OPA of $\frac{1}{f}$ can be. To address this question we used the Matlab command `fmincon` to minimize the quantity

$$(9) \quad \frac{{}_3F_2\left(\begin{smallmatrix} \alpha, \alpha, \frac{1}{2} \\ 1, 1 \end{smallmatrix}; \frac{4}{r^2}\right) + \frac{\alpha^2}{r^2} {}_3F_2\left(\begin{smallmatrix} \alpha+1, \alpha+1, \frac{3}{2} \\ 2, 3 \end{smallmatrix}; \frac{4}{r^2}\right)}{2\frac{\alpha}{r} {}_3F_2\left(\begin{smallmatrix} \alpha, \alpha+1, \frac{3}{2} \\ 2, 2 \end{smallmatrix}; \frac{4}{r^2}\right)}.$$

We find the minimum at the values $r = 2.533672086469380$, $\alpha = 2.551918826591946$ with the quotient equalling 0.975766335259681; see also the Mathematica graph below. For this value of r and α we find the degree one OPA of $\frac{1}{f}$ to equal

$$0.5018 - 0.2571(z_1 + z_2),$$

which, for instance, has a root at $z_1 = z_2 = 0.975766335259681$. This OPA has roots that are the closest (in $\|\cdot\|_1$, for instance) to the origin.



Graph of the quotient (9) where $r \in [2.4, 2.7]$ and $\alpha \in [2.4, 3]$.

2 Setup and proof intuition

The Hardy norm on $H^2(\mathbb{D}^2)$ is the ℓ_2 -norm of Taylor coefficients, equivalently the boundary $L^2(\mathbb{T}^2)$ -norm. The degree-one OPA for $1/f$ is the polynomial

$$pf = p_0 + p_1 z_1 + p_2 z_2$$

minimizing $\|1 - pf\|_{H^2}$.

The key is to view its normal equations through the probability measure

$$d\mu_f = \frac{|f|^2}{\|f\|_{H^2}^2} dm \quad \text{on } \mathbb{T}^2.$$

After dividing by p_0 , the two nonconstant normal equations depend only on the first moments of μ_f . If these measures converge to equal mass at two points $(1, 1)$ and (a, b) , the normalized OPA converges to the affine polynomial that vanishes at both points.

We choose a and b so that both rotate by about $2\pi/m$, but their relative rotation is only $2\pi/m^2$. The two interpolation rows are therefore almost proportional while asking for incompatible constant values. The affine coefficients grow like m , and its zero hyperplane passes within distance $O(1/m)$ of the origin.

3 OPA normal equations as moments

Lemma 1 (moment form and continuity). *Let $0 \neq f \in H^2(\mathbb{D}^2)$ with $f(0) \neq 0$, and let p_f be its degree-one OPA. Then $p_0 \neq 0$. Writing*

$$p_f = p_0(1 + q_1 z_1 + q_2 z_2),$$

the vector $q = (q_1, q_2)^T$ is the unique solution of

$$\left(\int_{\mathbb{T}^2} z_i \bar{z}_j d\mu_f \right)_{j,i=1}^2 q = - \left(\int_{\mathbb{T}^2} \bar{z}_j d\mu_f \right)_{j=1}^2. \quad (1)$$

If probability measures μ_ν converge weakly and the limiting 2×2 matrix in (1) is invertible, their normalized OPA coefficient vectors converge to the limiting solution.

Proof. The Gram matrix of $f, z_1 f, z_2 f$ is positive definite: an analytic identity

$$c_0 f + c_1 z_1 f + c_2 z_2 f = 0$$

forces the affine factor to vanish identically. The right side of the full normal system is $(\overline{f(0)}, 0, 0)^T$. Positivity and the Schur complement show that the first coordinate p_0 is nonzero.

For $j = 1, 2$, the right side is zero because $z_j f$ has zero constant term. Hence

$$0 = \langle p_f f, z_j f \rangle = p_0 \langle (1 + q_1 z_1 + q_2 z_2) f, z_j f \rangle.$$

Division by $p_0 \|f\|_{H^2}^2$ gives (1). Weak convergence gives convergence of all displayed continuous moments, and inversion is continuous at an invertible matrix. $\square$

4 A boundary concentration codebook

Fix an integer $m \geq 3$, put $D = m^2$, and let

$$\zeta = e^{2\pi i/D}.$$

Define the analytic polynomials

$$G_m(z_1, z_2) = z_1^m + z_1^{m-1} z_2^m, \quad (2)$$

$$L_\omega(z) = \frac{1}{D} \sum_{r=0}^{D-1} (z/\omega)^r, \quad (3)$$

$$H_m(z_1, z_2) = -1 + L_1(z_2) + L_{\zeta^{m+1}}(z_2). \quad (4)$$

Lemma 2 (two-point concentration). *For $t, M > 0$, set*

$$f_{m,t,M}(z) = \exp(tG_m(z) + MH_m(z)).$$

These functions are entire and zero-free. Their boundary probability measures satisfy

$$\lim_{M \rightarrow \infty} \lim_{t \rightarrow \infty} \mu_{f_{m,t,M}} = \frac{1}{2} \delta_{(1,1)} + \frac{1}{2} \delta_{(a_m, b_m)},$$

where

$$a_m = \zeta^m = e^{2\pi i/m}, \quad b_m = \zeta^{m+1} = e^{2\pi i(1/m+1/m^2)}.$$

Proof. On $\mathbb{T}^2$, $\operatorname{Re} G_m \leq 2$. Equality holds precisely when

$$z_1^m = 1, \quad z_1^{m-1} z_2^m = 1.$$

The exponent matrix

$$\begin{pmatrix} m & 0 \\ m-1 & m \end{pmatrix}$$

has determinant m^2 , so the maximum set consists of exactly

$$K_m = \{(\zeta^{mk}, \zeta^{(m+1)k}) : 0 \leq k < D\}.$$

All maxima are nondegenerate and have the same Hessian: in local angular coordinates the quadratic loss is

$$\frac{1}{2} [(mu)^2 + ((m-1)u + mv)^2].$$

Because $\gcd(m+1, D) = 1$, the second coordinates in K_m run through all D -th roots of unity. The root-of-unity filter L_ω equals one at ω and zero at the other such roots. Hence H_m equals zero at $(1, 1)$ and (a_m, b_m) , and equals -1 at every other point of K_m .

The elementary Laplace method at finitely many equal, nondegenerate maxima now gives, for fixed M , limiting masses proportional to $e^{2MH_m(x)}$. Thus the two selected points have weight one and each other point has weight e^{-2M} . Sending M to infinity proves the claim. $\square$

5 Exact limiting zero depths

Lemma 3. *For the equal two-point measure in Lemma 2, the limiting normalized OPA is*

$$\ell_m(z) = 1 + q_{1,m} z_1 + q_{2,m} z_2,$$

where

$$q_{1,m} = \frac{b_m - 1}{a_m - b_m}, \quad q_{2,m} = \frac{1 - a_m}{a_m - b_m}.$$

If $\rho_s(\ell) = \inf\{\|z\|_s : \ell(z) = 0\}$, then

$$\rho_s(\ell_m) = \frac{1}{\|(q_{1,m}, q_{2,m})\|_{s^*}},$$

where s^* is the dual exponent. In particular,

$$\rho_\infty(\ell_m) = \frac{\sin(\pi/m^2)}{\sin(\pi/m) + \sin(\pi/m + \pi/m^2)} \sim \frac{1}{2m}, \tag{5}$$

$$\rho_2(\ell_m) \sim \frac{1}{\sqrt{2}m}, \quad \rho_1(\ell_m) \sim \frac{1}{m}. \tag{6}$$

Proof. The two feature rows $(1, 1)$ and (a_m, b_m) are linearly independent because $a_m \neq b_m$. They therefore span all functions on the two-point space. The limiting normal equations say exactly that ℓ_m vanishes at both points:

$$1 + q_{1,m} + q_{2,m} = 0, \quad 1 + q_{1,m}a_m + q_{2,m}b_m = 0.$$

Solving gives the displayed coefficients.

Hölder's inequality, with equality in the dual direction, gives

$$\inf \{ \|z\|_s : q_{1,m}z_1 + q_{2,m}z_2 = -1 \} = \frac{1}{\|q_m\|_{s^*}}.$$

Finally,

$$\begin{aligned} |a_m - b_m| &= 2 \sin(\pi/m^2), \\ |1 - a_m| &= 2 \sin(\pi/m), \\ |1 - b_m| &= 2 \sin(\pi/m + \pi/m^2). \end{aligned}$$

Substitution proves (5); the ℓ_2 and ℓ_1 formulas follow by replacing the sum of the last two chord lengths by their Euclidean norm or maximum. Their asymptotics are immediate. $\square$

6 The full polynomial result

Theorem 4. *For every $\varepsilon > 0$ and each $s \in \{1, 2, \infty\}$, there exists a polynomial F with no zeros on $\overline{\mathbb{D}}^2$ such that its degree-one OPA p_F has a zero z with $\|z\|_s < \varepsilon$. Consequently,*

$$\inf_F \inf \{ \|z\|_s : p_F(z) = 0 \} = 0,$$

even when F ranges only over polynomials nonvanishing on the closed bidisk.

Proof. Choose m so large that the three quantities in (5)–(6) are below $\varepsilon/2$. By Lemmas 1 and 2, first choosing M large and then t large makes the normalized OPA coefficients of the zero-free entire function $f_{m,t,M}$ arbitrarily close to those of ℓ_m . The corresponding affine zero depths are continuous functions of the two nonconstant coefficients, so all three depths are below ε .

It remains to obtain a polynomial. The Taylor polynomials F_N of $f_{m,t,M}$ converge uniformly on the closed bidisk. Since the limiting entire function has a positive minimum modulus there, F_N is nonvanishing on the closed bidisk for all sufficiently large N . Uniform convergence also gives H^2 -convergence, hence convergence of the finite Gram matrices and of the degree-one OPA coefficients. Taking N large preserves the strict zero-depth inequalities. $\square$

7 Checks, upgrade attempts, and scope

Exact finite check. The packet-local script `code/verify_two_point_construction.py` checks the finite group relations, solves the two-point interpolation system, verifies the chord-length formula, and shows numerically the $1/m$ decay for all three norms. It is confirmatory only; every step is proved above.

Deep upgrades. The first attempt reduced the source’s symmetric two-parameter family to one-dimensional positive sums and reproduced its reported minimum 0.9757663352595. A second search over zero-free exponential families found explicit values below 0.15. The decisive upgrade replaced numerical search by the two-point concentration codebook, proving that the infimum is zero. A final upgrade used uniform Taylor approximation to obtain zero-free polynomials, rather than only zero-free entire functions.

Scope. The theorem completely answers the zero-depth extremal question for degree-one OPAs with the standard total-degree space $\text{span}\{1, z_1, z_2\}$, simultaneously for the three norms asked in the source. It does not address higher-degree OPAs, eventual escape of zeros, other monomial orderings, or Dirichlet-type spaces.

Novelty check. Bounded primary-source searches through 11 August 2026 covered the exact question, degree-one OPAs, zero-free polynomials on the bidisk, and zeros arbitrarily close to the origin. They recovered the two source papers and one-variable zero-exclusion results, but no primary source stating the theorem above. Novelty confidence is moderate because the search was bounded.

Human review. Audit the two-stage Laplace limit, the moment-to-interpolation passage, and the preservation of zero-freeness under uniform Taylor approximation.

References

- [1] C. Bénéteau, D. Khavinson, and D. Seco, *A counterexample to the weak Shanks conjecture*, Constr. Approx. 62 (2025), 645–661; [arXiv:2405.16943](#).
- [2] J. S. Geronimo and H. J. Woerdeman, *A Fourier analysis approach to disprove the weak Shanks conjecture*, Proc. Amer. Math. Soc. 154 (2026); [arXiv:2508.15938](#).